# Renwei Meng

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## EDUCATION

**Anhui University**, Hefei, China — Sep. 2023 – Jun. 2027
*B.Eng. in Digital Media Technology* — **GPA: 3.95/4.0**; **Rank: 1/100**; **IELTS: 7.0**
**Teaching Assistant:** Data Structures.
**Relevant Coursework:** Machine Learning, Probability and Statistics, Linear Algebra, Calculus I–III, Discrete Mathematics, Data Structures, Data Analysis, Operating Systems.

## RESEARCH INTERESTS

Reinforcement Learning for Foundation Models and Agents, with a focus on exploration, credit assignment, and reliable learning from feedback.

## PUBLICATIONS

**Peer-Reviewed Conference Publications**

- **FCR-PO: Learning Where to Retry in Language-Model Reasoning**.
  *First Author. ACML 2026.* **Accepted**.
- **Learning Which Evidence-Seeking Decisions Matter: Calibrated Counterfactual Replay for Language Agents**. *First Author. ACML 2026.* **Accepted**.
- **Know2Guess: A Contamination-Aware Multi-Zone Benchmark for Knowledge-Boundary Evaluation in Large Language Models**. *First Author. ICONIP 2026.* **Accepted**.
- **Conditional Riemannian Adaptive Flow Transport for Cross-Subject EEG Decoding**.
  *First Author. PRICAI 2026.* **Accepted**.
- **Group Resonance Network: Learnable Prototypes and Multi-subject Resonance for EEG Emotion Recognition**. *First Author. ICANN 2026.* **Published**.
- **S-AODP++: Structure-Aware AOI-Guided Scanpath Attention for AI-Generated Image Identification**. *Co-first Author. ICMI 2026.* **Published**.
- **AuRA: Aligning Voice and Physiology by Autonomic Self-Supervision for Label-Efficient Affective and Clinical Sensing**. *First Author. BIBM 2026.* **Accepted**.
- **CausalG2C: An Intervention-Aware Genome-to-Cell Model for Zero-Shot Genetic Perturbation Prediction**. *First Author. BIBM 2026.* **Accepted**.

**Manuscripts Under Review**

- **Normal Fréchet Drift: Measuring Post-Training Departure from Pretrained Reference Manifold**.
  *Co-first Author. ICLR 2027.* **Under Review**.

**Preprints**

- **VulnAgent-R2: Evidence-Calibrated Multi-Agent Auditing for Repository-Level Vulnerability Detection**. *Co-first Author.* **arXiv Preprint**.
- **Explainable Innovation Engine: Dual-Tree Agent-RAG with Methods-as-Nodes and Verifiable Write-Back**. *First Author.* **arXiv Preprint**.

## RESEARCH EXPERIENCE

**Brain-Informed Distillation for Reasoning and World Models** — Oct. 2026 – Present
*Lead Researcher | Advisor: Zhi Lu | Tsinghua University*

- Developing human brain distillation methods that use EEG and fMRI as auxiliary training signals to improve model reasoning and physical prediction, without requiring brain recordings at inference.
- Developed **RELAY** to distill temporal EEG structure into visual reasoning; improved held-out family accuracy from **62.75% to 71.47%**, without brain recordings at inference.
- Developed **SCOPE**, using EEG/fMRI-calibrated forecast losses to improve world models; raised Physion contact accuracy from **70.58% to 78.62%** and PushT success from **62.50% to 70.40%**.

**Generative Model Post-Training and Reward-Hacking Diagnostics** — Jul. 2026 – Oct. 2026
*Lead Researcher | Advisor: Prof. Tao Lin | Westlake University*

- Proposed **Normal Fréchet Drift (NFD)**, a geometric metric for assessing reward hacking by quantifying normal deviations from a pretrained reference manifold, distinct from tangential redistribution.
- Constructed bounded local tangent patches from pretrained samples; analyzed reconstruction bias through coverage, curvature, tangent, and anchor errors, separately from evaluation sampling variance.
- Evaluated controlled geometries and SD3.5-M / FLUX.2 checkpoints, checking robustness across encoders, prompt splits, and reference refits to assess post-training and distillation effects.

## RESEARCH EXPERIENCE (CONTINUED)

**Reliable Reinforcement Learning for Language Models and Agents** Jan. 2026 – Jul. 2026
*Independent Research*

- Developed **FCR-PO**: compare predicted success of ordinary continuation and targeted intervention at the same prefix, allocate a bounded retry budget, and train on verified repairs while retaining prior generation behavior.
- Achieved **77.70%** mean accuracy across MGSM, ProofWriter, and Spider versus **67.83% for GRPO**, using Qwen3-8B-Base at the same 300M-token training budget; tested repair selection with matched-record controls.
- Developed **PCCC-RL** for tool-using agents: compare actions from restored tool states, correct predicted credit with randomized replay, and allocate replay effort by uncertainty, learning sensitivity, and execution cost.
- Improved joint answer–evidence exact match over an access-matched replay baseline by **4.19 / 3.86 points** on HotpotQA / MuSiQue at a shared 100M-token budget, averaged over eight seeds.

**Multimodal Learning and AI for Science** Sep. 2025 – Jul. 2026
*Lead Researcher | Advisor: Prof. Shiang Hu | Anhui University*

- Developed **AuRA**: self-supervised voice–physiology alignment with temporal optimal transport and distillation for label-efficient, voice-only sensing.
- Developed **CausalG2C**: intervention-aware regulatory graphs and latent flow matching for zero-shot prediction of gene-perturbation responses.

**Evidence-Calibrated Agents for Software Vulnerability Detection** Mar. 2025 – Jan. 2026
*Lead Researcher | Advisor: Chen Zhang | Anhui University*
*National College Student Innovation Program*

- Built code-structure models using AST / CFG representations and curated C/C++ evaluation data; extended this work into **VulnAgent-R2** with repository-level evidence gathering and role-specialized agents.
- Developed counter-evidence and selective verification mechanisms for grounded findings; evaluated detection, localization, and cost, reducing online tokens by **38.3%** versus always-full multi-agent execution on JITVul.

## HONORS & AWARDS

- **National Scholarship** 2025
- **Finalist**, Mathematical Contest in Modeling (MCM/ICM) 2026
- **National Second Prize**, National College Students Market Survey and Analysis Competition 2026
- **First Prize**, National College Student Mathematical Modeling Competition (Provincial) 2025

## TECHNICAL SKILLS

**Programming:** Python, C/C++, MATLAB, Java, R, SQL, Bash / zsh.
**Machine Learning & Agents:** PyTorch, TensorFlow, LangChain, LangGraph.
**Research Infrastructure:** Linux, Git, Docker, LaTeX.

## ADDITIONAL EXPERIENCE

- **Java Backend Developer Intern**, Taiyuan Longwei Electronic Technology Co., Ltd. Jul. – Sep. 2025
  Built Spring Boot / MyBatis services and RESTful APIs; coordinated database and frontend integration.
- **Systems Operations Intern**, State Grid Shanxi Electric Power Co., Ltd. Jul. – Sep. 2024
  Maintained power-system services and analyzed operational metrics for reliability reports.
- **Coronary CTA Artifact Removal** — Lead Researcher, Anhui University. Jan. – Jun. 2025
  *Advisor: Prof. Zhe Jin*
  Combined U-Net and GAN denoising to improve SNR and SSIM over conventional filters.
- **Carbon Price Prediction** — Lead Researcher, Anhui University. Jun. 2024 – Jun. 2025
  *Advisor: Prof. Zhang-HangJian Chen*
  Developed a CNN–recurrent attention model across seven Chinese carbon markets.

## PATENTS

- **A generative deepfake video detection method and its system**. Co-inventor.
  Chinese invention patent application No. 202610165744.1.
- **Resting-state EEG quality assessment based on dual-branch contrastive learning**. Co-inventor.
  Chinese invention patent application No. 202512044739.3.